

Chapter 4

Buying Power Reduction

Having discussed the nature of implied volatility (IV) and the general risk-reward profile of short premium positions, it's time to introduce some elements of short volatility trading in practice. Because short options are subject to significant tail risk, brokers must reserve a certain amount of capital to cover the potential losses of each position. The capital required to place and maintain a short premium trade is called the buying power reduction (BPR), and the total amount of portfolio capital available for trading is the portfolio buying power.

BPR is the amount of capital required to be set aside in the account to insure a short option position, similar to escrow. BPR is used to evaluate short premium risk on a trade-by-trade basis in two ways:

1. BPR acts as a fairly reliable metric for the worst-case loss for an undefined risk position in current market conditions.
2. BPR is used to determine if a position is appropriate for a portfolio with a certain buying power.

Though BPR is the option counterpart of stock margin, the distinction between the two *cannot be overstated*, as short options positions can never be traded with borrowed money. BPR is *not* borrowed money nor does it accrue interest. It is *your* capital that is out of play for the duration of the short option trade. Margin, mostly used for stock trading, is money borrowed from brokers to purchase stock valued beyond the cash in an account. Interest *does* accrue on margin (usually between a 5% to 7% annual rate), and traders are required to pay back the margin plus interest regardless of whether the stock trade was profitable. Margin and BPR are conceptually different: Margin amplifies stock purchasing power, and BPR lowers purchasing power to account for the additional risk of short options.

The definition of BPR and its usage differs depending on whether the strategy is long or short and whether the strategy has defined or undefined risk. For long options, the maximum loss is simply the cost of the option, so

that is the BPR. Defining the BPR for short options is more complicated, particularly for undefined risk positions, because the loss is theoretically unlimited. Defined risk trades, which will be covered in the next chapter, are short premium trades with a known maximum loss. These are simply short premium contracts (undefined risk trades) combined with cheaper, long premium contracts that will cap the excess losses when the underlying price moves past the further strike. BPR is the maximum loss for a defined risk strategy, but only an estimate for maximum loss for an undefined risk trade. Because the undefined risk case is more complicated, this chapter explains the BPR as it relates to undefined risk strategies, specifically short strangles.

Up until now, options trading has predominantly been discussed within the context of strangles, an undefined risk strategy with limited gain and theoretically unlimited loss. In this case, the BPR is calculated such that it is unlikely that the loss of a position will exceed that threshold. More specifically, BPR is intended to account for roughly 95% of potential losses with exchange-traded fund (ETF) underlyings and 90% of potential losses with stock underlyings.¹ The historical effectiveness of BPR for an ETF underlying is seen in [Figure 4.1](#) by looking at losses for 45 days to expiration (DTE) 16  SPY strangle from 2005–2021.

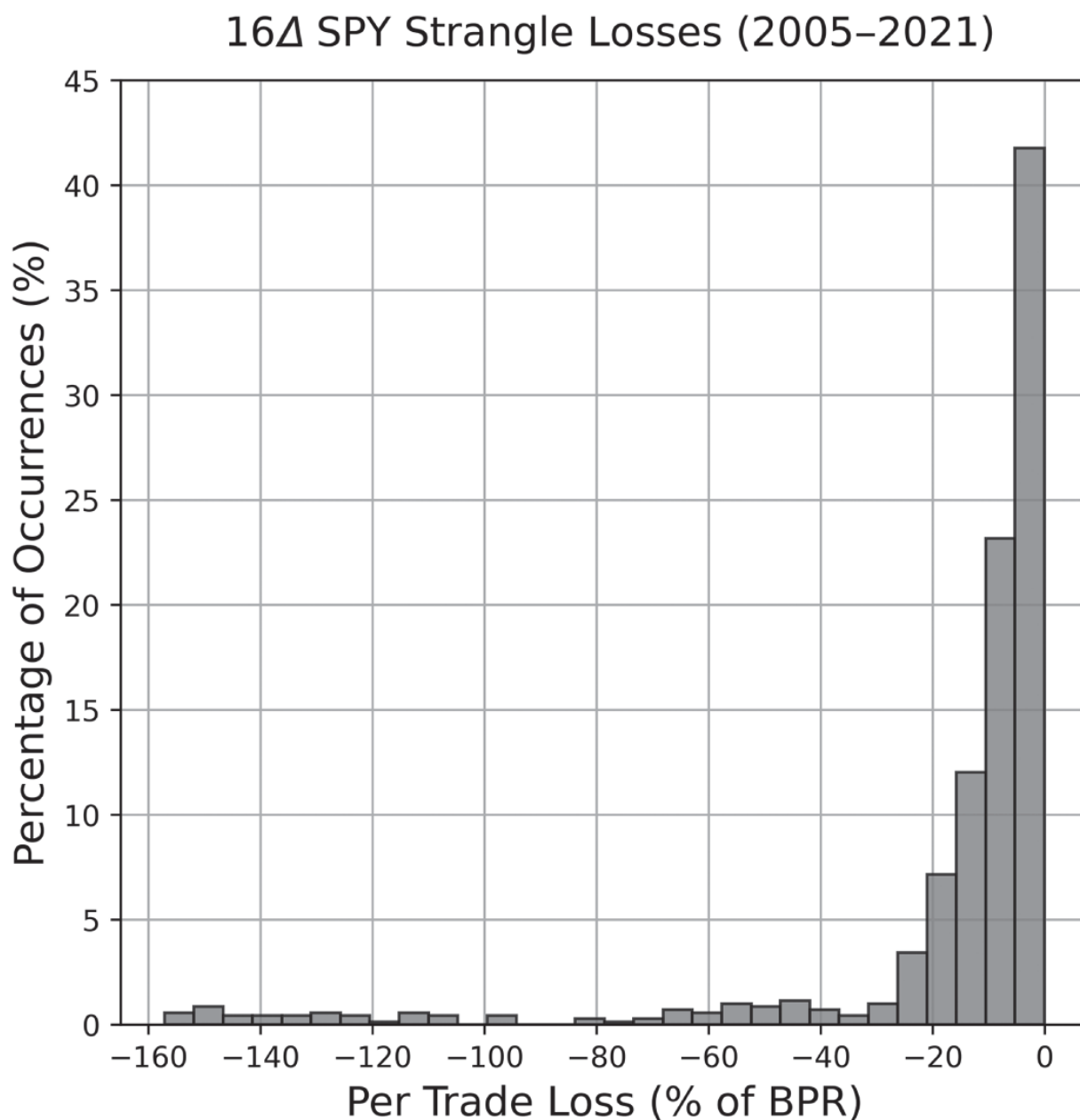


Figure 4.1 Loss as a % of BPR for 45 DTE 16 Δ SPY strangles held to expiration from 2005–2021.

In this example, most losses ranged from 0% to 20% of the BPR. Roughly 95% of all these losses were accounted for by the BPR when this position was held to expiration, as expected. Though BPR did not always capture the full extent of realized losses, it is an effective proxy for worst-case loss on a trade-by-trade basis in most cases. This metric works fairly well for SPY strangles, but strangles with more volatile underlyings and strangles with

tighter strikes may be more likely to have losses that breach BPR (hence the 90% efficacy rate for stocks).

BPR corresponds to the capital required to place a trade, and that quantity varies depending on the specific strategy. The BPR for short strangles can be approximated as 20% of the price of the underlying, but mathematically, BPR depends on three variables: the stock price, put/call prices, and the put/call strike prices.² Because the strangle is composed of the short out-of-the-money (OTM) call and short OTM put, the BPR required to sell a strangle is simply the larger of the short put BPR and the short call BPR. The short call/put BPR is the largest of three different values:

1. $((0.2 \times \text{stock price}) - |\text{strike price} - \text{stock price}|) \times 100$, which is the expected loss from a 20% move in the underlying price.
2. $0.1 \times \text{strike price} \times 100$, which is the expected loss from a 10% strike breach.
3. $250 - \text{contract price}$, which ensures that there is a minimum BPR for cheap options.

As BPR is intended to encompass the largest likely loss for an undefined risk contract, the largest of these values is taken. This can be mathematically represented using the *max* function, which takes the largest of the given values:

$$\begin{aligned} \text{Short Put BPR} = \max & (((0.2 \times S) - (S - K)) \quad (4.1) \\ & \times 100, (0.1 \times K) \\ & \times 100, 250 - P \times 100) \end{aligned}$$

$$\begin{aligned} \text{Short Call BPR} = \max & (((0.2 \times S) - (K - S)) \quad (4.2) \\ & \times 100, (0.1 \times K) \\ & \times 100, 250 - C \times 100) \end{aligned}$$

Combining these formulas, the BPR of the strangle is given by:

$$\text{Short Strangle BPR} = \max(\text{Put BPR}, \text{Call BPR}) \quad (4.3)$$

Clearly, this equation is hairy, but using some numerical examples, one can infer how strangle BPR and, therefore, option risk changes with more intuitive variables, such as the historical and implied volatility of the underlying. Consider three potential strangle trades outlined in [Table 4.1](#).

Table 4.1 Three examples of approximate 45 DTE 16 Δ strangle trades with different parameters and the resulting BPR.

	Scenario A	Scenario B	Scenario C
Stock Price	\$150	\$150	\$300
Call Strike	\$160	\$175	\$320
Put Strike	\$140	\$130	\$280
Call Price	\$1	\$2	\$2
Put Price	\$1	\$2	\$2
BPR	\$2,000	\$1,750	\$4,000
IV	20%	45%	20%

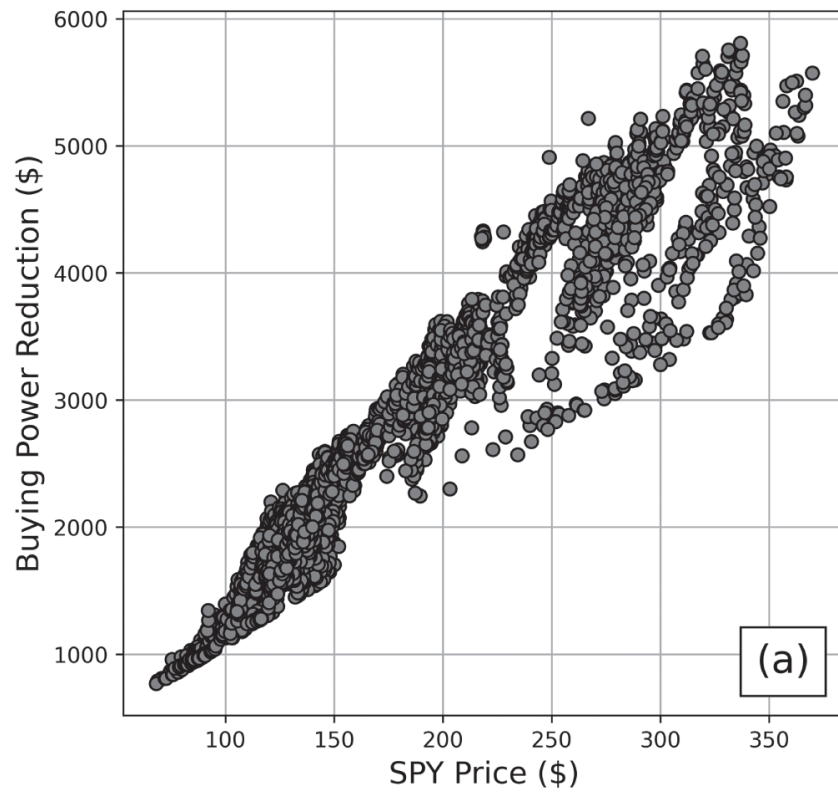
The underlying in Scenario B is priced the same as that of Scenario A, but the strikes for the 16 Δ strangle are further apart (consistent with a higher implied volatility). The underlying in Scenario C is twice as expensive as the underlyings in Scenarios A and B, but the IV in Scenario C is the same as that of Scenario A.

Because the BPR is higher in Scenario C compared to Scenario A (but the implied volatility and contract delta are the same), traders can deduce that strangle BPR tends to increase with the price of the underlying. *Technically*, BPR is inversely correlated with option price, but the BPR still tends to increase with the price of the underlying because more expensive instruments have larger volatilities (as a dollar amount) and, therefore, higher potential losses. BPR also decreases as the IV of the underlying

increases, and both relationships can be seen in [Figure 4.2](#) looking at BPR for 45 DTE 16 Δ SPY strangles from 2005–2021.

These charts show a strong linear relationship between BPR and underlying price and a slightly messier inverse relationship between BPR and underlying IV. This relationship is largely driven by the strikes moving further OTM for a fixed Δ as IV increases. BPR tends to decrease exponentially as the IV of the underlying increases, and because BPR is a rough estimate for worst-case loss, this relationship illustrates how the magnitude of potential outlier losses tends to decrease when IV increases.³

SPY Price and 16Δ Strangle BPR (2005–2021)



VIX and 16Δ Strangle BPR (2005–2021)

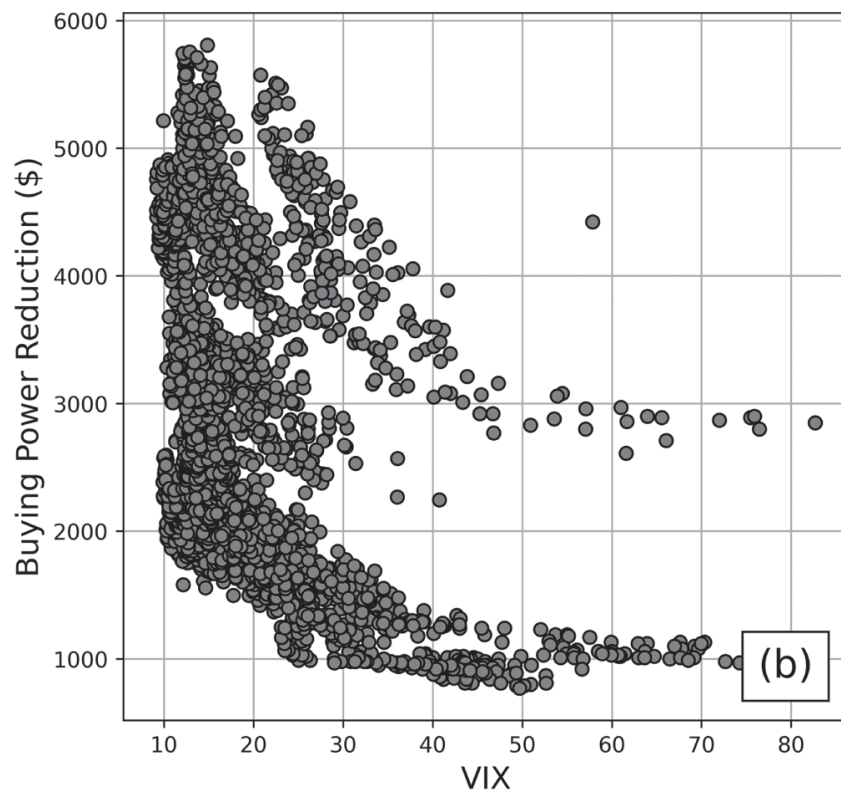


Figure 4.2 Data from 45 DTE 16 Δ SPY strangles from 2005–2021. (a) BPR as a function of underlying price. (b) BPR as a function of underlying IV.

Short premium positions carry higher credits and larger profit potentials when IV is high, but the reduction in BPR also allows more short premium positions to be placed compared to when IV is low. Because average profit and loss (P/L) is higher on a trade-by-trade basis *and* more potentially profitable positions can be opened, it is essential to reserve a large percentage of portfolio buying power for high IV conditions. This additionally justifies increasing the percentage of portfolio capital allocated to short premium BPR as IV increases. These crucial high-IV profits improve portfolio performance but also cushion potential future losses. Historically, when the VIX has been over 40 compared to under 15, the same amount of capital has covered the BPR of roughly twice as many 16 Δ SPY strangles. The difference between the number of short premium trades allowed in these two volatility environments is even larger when taking portfolio allocation guidelines into account. For context, consider the scenarios outlined in [Table 4.2](#).

Table 4.2 Two portfolios with the same net liquidity but different amounts of market volatility, using SPY strangle data from 2005–2021.

	Scenario A	Scenario B
Net Portfolio Liquidity	\$100,000	\$100,000
Current VIX	> 40	< 15
Portfolio Allocation	\$50,000	\$25,000
Approx. 16 Δ SPY Strangle BPR	\$1,500	\$3,300
Max Number of Strangles	33	7

It's important to note that BPR can be used to compare the capital at risk for variations of the same type of strategy, but it *cannot* be used to compare the risk between defined risk strategies and undefined risk strategies. For example, if the BPR required to trade a short strangle with underlying A was twice the BPR required to trade a short strangle with underlying B and otherwise had identical parameters, we can conclude that A is twice as risky

as B. This is a valid comparison because we are considering two trades with the same risk profile, but BPR *cannot* be used to compare strategies with different risk profiles (say, a short strangle versus a short put) because it does not account for factors like the probability of profit or the probability of incurring a large loss. This subtlety will be discussed in more detail in the following chapter.

Understanding BPR is crucial when transitioning from options theory to applied options trading because it corresponds to the actual capital requirements of trading short options. BPR is also necessary to discuss the capital efficiency of options (option leverage) in entirety. Consider a stock trading at \$100 with a volatility of 20%, and suppose a trader wanted to invest in this asset with a bullish directional assumption. The trader could achieve a bullish directional exposure to this underlying in a few different ways as shown with the examples in [Table 4.3](#).

Table 4.3 Example trades that offer bullish directional exposure. Assume that the 50 Δ (ATM) call and put contracts have 45 DTE durations and cover 100 shares of stock.

Strategy	Capital Required	Max Profit	Max Loss	Probability of Profit (POP)
50 Shares of Long Stock	\$5,000	∞	\$5,000	50%
Long 50 Δ Call	\$280	∞	\$280	30%
Short 50 Δ Put	\$2,000 (BPR)	\$280	\$9,720	60%

In this one-to-one comparison, the effects of option leverage are clear because the long call position achieves the same profit potential as the long stock position with 94% less capital at risk. The short put position is capable of losing several times the initial investment of the trade but has a higher POP than the long stock position and requires 60% less capital. Suppose that the price of the stock increases to \$105 after 45 days. The resulting profits and corresponding returns for these different positions is given below:

- Long stock:
 $\text{Profit} = 50 \text{ shares} \times (\$105 - \$100) = \250
- Long ATM call:
 $\text{Profit} = 100 \text{ shares} \times (\$105 - \$100) - \$280 = \$220$
- Short ATM put: $\text{Profit} = \$280$

In this example, the long call position was able to achieve 88% of the long stock profit with 94% less capital, and the short put position was able to achieve 12% *more* profit than the long stock position with 60% less capital.

Takeaways

1. Because short premiums are subject to significant tail risk, brokers must reserve capital to cover the potential losses of each position. This capital is called BPR. The total amount of portfolio capital available for trading is called portfolio buying power.
2. BPR is used to evaluate short premium risk on a trade-by-trade basis in two ways: BPR is a fairly reliable metric for worst-case loss of an undefined risk position, and BPR is used to determine if a position is appropriate for a portfolio based on its buying power.
3. For long options, BPR is the cost of the option. For short strangles, the BPR is roughly 20% of the price of the underlying. BPR for short options encompasses roughly 95% of potential losses for ETF underlyings and 90% of losses for stock underlyings.
4. Strangle BPR tends to increase linearly with the price of the underlying because more expensive instruments have larger volatilities (as a dollar amount) and, therefore, higher potential losses. There is an inverse relationship between strangle BPR and underlying IV; more specifically, it approximately decreases exponentially as the IV of the underlying increases. This demonstrates the advantages of trading short when IV is high because more short strangles can be opened with the same amount of capital as in low IV, and the outlier loss potential is generally lower.

5. BPR can be used to compare capital at risk for variations of the same strategy, but it cannot be used to compare the risk of different strategies with different risk profiles.
6. The leveraged nature of options allows traders to achieve a desired risk-return profile with significantly less capital than an equivalent stock position.

Notes

- 1 This statistic will vary with the IV of the underlying, but this is a suitable approximation for general cases.
- 2 This is the FINRA (Financial Industry Regulatory Authority) regulatory minimum. Brokers typically follow this formula, but occasionally (especially when IV is very high) they will increase the capital requirements for contracts on specific underlyings.
- 3 This relationship between BPR and IV is specific to strangles. The next chapter discusses how these relationships may differ for certain defined risk strategies.